

Configuration options used

Table 1 discusses some of the common configuration options used in this article.
Table 1: Configuration options

Option	Purpose
<i>-P</i>	Specifies on which port DeleGate will listen for requests from clients. You can specify a host or IP address too, such as <i>-P local-host:8080</i> or <i>-P 192.168.1.1:8080</i> . By default, the port is a TCP port, unless a UDP-based application protocol is specified in the <i>SERVER</i> parameter. However, you can still specify UDP—for example, <i>-P8080/UDP</i> . You can also specify multiple host:port options as a comma-separated list.
<i>-T</i>	Used to trace system calls. Signals occurring in the DeleGate process will be watched by the parent DeleGate using <i>"ptrace(2)"</i> . <i>-Tx</i> here ensures that DeleGate processes trying to execute the <i>"execve(2)"</i> system call will be trapped and killed. This is a very useful security measure.
<i>DGROOT</i>	At startup, DeleGate searches for an available DGROOT which is both readable and writable by the OWNER. The OWNER (which can be set as a variable also) is the user, using which the delegate process will run. By default this user is the user that launched the delegate process. In our example, this is the 'delegate' user we created in the previous section. Typically it will be <i>\$(HOME)/delegate</i> and <i>"/"</i> if CHROOT is used. This parameter needs to be specified on the command line.
<i>SERVER</i>	Specifies the protocol to be used for communication with clients.
<i>REMITTABLE</i>	Only the listed protocols (to the <i>SERVER</i>) will be permitted to be relayed by the proxy. A typical list of protocols expected to be relayed through an HTTP proxy includes HTTP, HTTPS, Gopher, FTP and WAIS.
<i>CACHE</i>	Whether the proxy should use cache or not. A value of 'no' will disable cache; 'do' will enable it, and create <i>CACHEDIR</i> if it does not exist.
<i>TIMEOUT</i>	The time-out period for an action, by default, is in units of seconds. A '0' means 'never time out'. Valid units are days, hours and minutes (e.g., 1d, 1h, 1m). The action can be, for example, <i>'shutout'</i> which specifies that a suspicious client host will be shut out until a relevant file (under <i>ADMDIR/shutout/</i>) is removed, or the file is expired by <i>TIMEOUT</i> . This is a good defence against attacks on the proxy.
<i>MAXIMA</i>	Maximum values for resource usage, processes, connections etc. <i>'randstack'</i> randomises the range of stack base for security, <i>'randenv'</i> randomises the range of environment variables base and <i>'randfd'</i> randomises the range of client socket file-descriptors.

<i>ADMIN</i>	Specifies the admin contact person who will receive e-mails from DeleGate on various events.
<i>AUTHORIZER</i>	Specifies an optional server for authentication and authorisation. A list of user names and passwords is used in pairs for authentication. There are various other schemes available, such as PAM.

DeleGate as an HTTP proxy

In this sample configuration, DeleGate is configured as an HTTP proxy with authentication and no caching. To enable caching, change *"CACHE=no"* to *"CACHE=do"*:

```
# su - delegate
$ vi /opt/delegate/etc/dg.conf
-P8080
-Tx
DGROOT="/opt/delegate"
SERVER=http
REMITTABLE="http,https/443,gopher,ftp,wais"
CACHE=no
TIMEOUT="shutout:30m"
MAXIMA="randstack:32"
MAXIMA="randenv:1024"
MAXIMA="randfd:32"
ADMIN=tech.support@unixclinic.net
AUTHORIZER="-list{user1:password1,user2:password2}"
```

Moving on

This article is only an introduction to this marvellous piece of engineering. We have not covered many DeleGate features such as SOCKS proxy, transparent SOCKS proxy, chaining of proxies, etc. I leave further exploration to readers; I am pretty sure that those of you who are interested will explore and benefit from DeleGate's capabilities. Interested readers could also explore another proxy server—Dante (the link is in the *References* section below). Time permitting, I may, in the future, write about implementing other features of DeleGate. Dante is also in my to-do list, so watch out. 

References

- [1] DeleGate home page: <http://www.delegate.org/delegate/>
- [2] Dante - A Free SOCKS Proxy Implementation: <http://www.inet.no/dante/index.html>

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